CHLORPHENAMINE

Sedating

Anaphylaxis adjunct

CHEMICAL NAME (RS)-3-(4-chlorophenyl)-N,N-dimethyl-3-(pyridin-2-yl)propan-1-amine (maleate); also chlorpheniramine

CLASS First-generation (sedating) H1 antihistamine (alkylamine)

TRADE NAMES Piriton · Chlor-Trimeton · Allergex · Histafen

THERAPEUTIC USES / INDICATIONS

- Allergic reactions: urticaria, angioedema, allergic rhinitis, insect bites.
- Anaphylaxis (adjunct after adrenaline — for itch/urticaria).
- Blood-product transfusion reactions; prevention of drug reactions (premedication).
- Pruritus (chickenpox, eczema).

ADMINISTRATION

- **PO** tablets or syrup.
- **IV** slowly over 1 minute (rapid → CNS stimulation, hypotension); **IM**.
- In anaphylaxis: only after IM adrenaline.

DOSAGES

- **Adult PO:** 4 mg every 4–6 h; max 24 mg/day.
- **Adult IV/IM:** 10 mg (repeat up to 40 mg/24 h).
- **Children PO:** 1–2 yr 1 mg twice daily; 2–5 yr 1 mg every 4–6 h; 6–11 yr 2 mg every 4–6 h.
- **Children IV/IM:** 6 months–6 yr 2.5 mg; 6–12 yr 5 mg.

PHARMACOKINETICS

- **Onset:** PO 30–60 min; IV minutes. **Duration:** 4–6 h.
- Metabolised in liver.
- Excreted in urine. **t½:** 12–24 h.

NURSING MANAGEMENT

- Give IV slowly; monitor BP.
- Assess itch, rash, swelling.
- Safety: falls, no driving.
- In anaphylaxis, ensure adrenaline given first.

PHARMACODYNAMICS

- Competitive **H1-receptor antagonist** → ↓ itching, wheal, flare, rhinorrhoea, capillary permeability.
- Crosses blood–brain barrier → sedation; anticholinergic effects.

SIDE & ADVERSE EFFECTS

SIDE EFFECTS (COMMON)

- Drowsiness, impaired concentration.
- Dry mouth, blurred vision.
- Dizziness.

ADVERSE EFFECTS (SERIOUS)

- Paradoxical excitation in children; seizures (overdose).
- Urinary retention; angle-closure glaucoma.
- Hypotension (rapid IV).
- Confusion in elderly.

CONTRAINDICATIONS & CAUTIONS

- Hypersensitivity.
- Neonates and premature infants; MAOI use within 14 days.
- **Caution:** glaucoma, prostatic hypertrophy, epilepsy, elderly, driving.

KEY INTERACTIONS

- **Alcohol, opioids, benzodiazepines:** ↑ sedation.
- **Anticholinergics, TCAs:** ↑ antimuscarinic effects.
- **Phenytoin:** ↑ levels.

PATIENT EDUCATION

- May cause drowsiness; do not drive or use machines.
- Avoid alcohol.
- Children may become excitable.

KEY POINTS

- Sedating H1 antihistamine.
- IV slowly over 1 min.
- Adjunct (not first-line) in anaphylaxis.
- Drowsiness and anticholinergic effects.

SOURCES CHECKED **[476]** Chlorphenamine 10 mg/ml Solution for injection (UK SmPC) **[477]** 21 CFR Part 341 (US OTC monograph)

Full references p. 432–457